Homework

1. Download the daily closing prices for Microsoft (MSFT), Apple (AAPL), and Google (GOOGL) from Yahoo Finance over the period 05/23/2024–05/22/2026. Report the first five and last five observations in the dataset, and explicitly state the total number of observations.

2. As of 05/22/2026, the market capitalizations were as follows: Microsoft, \$3.109 trillion; Apple, \$4.536 trillion; and Google, \$4.64 trillion. The objective is to compute the Value at Risk (VaR) of a market-capitalization-weighted portfolio of these three stocks. The portfolio size is \$1,000,000, and the risk-factor changes are the log returns X.

   For the market-capitalization-weighted portfolio

   $$V_t = \sum_{i=1}^{3} \lambda_t^{(i)} S_t^{(i)},$$

   determine the market weights

   $$\omega_i = \frac{\lambda_t^{(i)} S_t^{(i)}}{V_t}.$$

3. Using the empirical distribution of the log returns, estimate the VaR of the market-capitalization-weighted portfolio at the 95% confidence level under the loss operators $l_{[t]}$, $l_{[t]}^{\text{lin}}$, and $l_{[t]}^{\text{quad}}$.

4. Estimate the mean vector and covariance matrix of the daily log returns using the exponentially weighted moving average (EWMA) method. Start from an initial pair μ_0, Σ_0; for example, it is acceptable to set both equal to 0. Then update according to the EWMA recursions:

   $$\mu_{t+\Delta} = \lambda \mu_t + (1-\lambda) X_t$$
   $$\Sigma_{t+\Delta} = \theta \Sigma_t + (1-\theta)(X_t - \mu_t)(X_t - \mu_t)^T.$$

   This yields estimates for μ and Σ as of Friday, 05/22/2026, which are then to be used over the subsequent period 05/22/2026–05/25/2026. Use $\lambda = \theta = 0.97$. Note that there is no trading on 05/23/2026 and 05/24/2026, so the next trading day after 05/22/2026 is 05/25/2026.

5. Estimate the VaR of the market-capitalization-weighted portfolio at the 95% confidence level under the assumption that the log returns are normally distributed with the estimated mean vector and covariance matrix. Use the loss operators $l_{[t]}$, $l_{[t]}^{\text{lin}}$, and $l_{[t]}^{\text{quad}}$. For the full and quadratic loss operators, use MC simulation to obtain the VaR by sampling the corresponding normal random vectors. For the linear loss operator, obtain the VaR exactly using the analytical method. In the simulation, use $N = 100,000$ samples.

6. Compare the resulting VaR estimates. Determine whether one or more estimates differ materially from the others, and, if so, explain the likely reason.